2023-Winter Paper 4 – Recent Advances

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Q1. Laproendoscopic single site surgery: LESS

Type: LAQ (25 marks)

Read as laparoendoscopic single-site surgery (LESS). NOTES (natural orifice transluminal endoscopic surgery) is covered in 2022-Summer Paper 4, Q1.

Definition and concept

- **LESS:** laparoscopic or robotic surgery in which all instruments and the camera enter through **one skin incision**, usually hidden in the umbilicus or below the waistline.
- Aim: further reduce access trauma, giving **better cosmesis, less pain and shorter stay**. Chapter 142 places it on a spectrum: multiport → hand-assisted → LESS → SP (single-port) robot → NOTES. Each step to fewer incisions buys cosmesis at the price of working space and triangulation.
- First urological single-port nephrectomies were reported in 2007; the first single-port robotic urological surgery followed in 2009.

Nomenclature (LESSCAR)

- Early names were many: SILS (single-incision laparoscopic surgery), SPA (single-port access), OPUS (one-port umbilical surgery), TUES (transumbilical endoscopic surgery), e-NOTES (embryonic NOTES).
- **LESSCAR (Laparoendoscopic Single-Site Surgery Consortium for Assessment and Research, 2008)** agreed the single umbrella term **LESS** and set standards for reporting, training and research.
- Sub-types: transumbilical or extraumbilical; **pure LESS**; **"LESS + 1"** or needlescopic-assisted (an extra 2–5-mm port); **R-LESS** (robotic LESS).

Access: ports and devices

Option	Description
Several low-profile trocars	Placed close together through one fascial incision (the future extraction incision)
Multichannel ports	TriPort 15 (two 5-mm + one 15-mm); TriPort+ (three 5-mm + one 10-mm); SILS port; GelPOINT (gel cap, trocars placed anywhere on it)
da Vinci SP entry guide	One 25–27-mm cannula: three articulating 6-mm instruments + one flexible articulating camera

- Position: modified flank (transperitoneal) or full flank (retroperitoneal SP, 2.5–3-cm incision near the 12th rib tip, balloon dilatation).

The triangulation problem

- In standard laparoscopy ports are spread so instruments meet the target at an angle (the "diamond"). In LESS all shafts are **parallel and in-line** with the camera.
- Results: **instrument clashing**, **"crossing swords"**, poor depth perception, poor retraction and hard suturing.
- Solutions:
  - **Articulating instruments** (tip bends inside the abdomen) and **pre-bent or curved instruments**: they cross at the fascia, so the right hand works on the left side (**chiral manipulation**).
  - **Flexible-tip or 5-mm low-profile scopes**, offset optics, extra-long instruments and instruments of **different lengths** to avoid handle clash.
  - A 2-mm needlescopic instrument or extra 5-mm port without hesitation.
  - **Robotic platforms** restore wristed movement.

Robotic single-site surgery

Platform	Features
da Vinci Single-Site (Si/Xi)	Curved crossing cannulas; semi-rigid **non-wristed** instruments; software swaps hand control
da Vinci SP (FDA [Food and Drug Administration] approval 2018)	Purpose-built; one arm, no external collisions; instruments with elbow and wrist; 3D articulating camera; on-screen navigator shows instrument positions

- SP needs **10–25 cm between cannula and target** to deploy the joints; a GelPort lets instruments "float" outside the skin. Assistance through an **"SP + 1" port** or ROSI (remote-operated suction-irrigation). Limited by cost and availability.

Indications

- **Kidney:** simple and **donor nephrectomy**, radical nephrectomy, partial nephrectomy, **pyeloplasty**, nephroureterectomy, cyst decortication, cryoablation.
- **Adrenalectomy**; varicocelectomy; orchidopexy and pyeloplasty in children.
- **Pelvis:** SP radical prostatectomy (transperitoneal, extraperitoneal); **SP transvesical simple prostatectomy** and bladder diverticulectomy.
- **Best case:** thin patient, small or benign lesion, experienced surgeon.
- **Relative contraindications:** obesity, large or hilar tumours, complex vascular anatomy, dense adhesions, unfit for pneumoperitoneum.

Outcomes versus conventional laparoscopy (Chapter 142)

- **Randomised trial:** same operative time, blood loss, complications and stay; **less analgesia**.
- **Donor nephrectomy meta-analysis:** same stay, recovery, transfusion and conversion; **lower pain scores**.
- **Renal mass:** same perioperative outcome and survival.
- **SP robotic partial nephrectomy vs multiport:** shorter stay, less opioid; same margins and complications; **longer warm ischaemia** in one review.
- An **extra port is often needed**; conversion is mostly to laparoscopy, rarely open.
- **Net effect:** gains are confined to **pain, cosmesis and stay**; oncological and functional outcomes are unchanged, but the operation is technically harder and longer.

Complications

- All laparoscopic complications (access injury, vascular and bowel injury, CO_2 effects).
- **Incisional and umbilical hernia** is higher (larger fascial defect): about 5% with da Vinci Xi single-site, lower with SP.
- Wound infection; conversion; longer operation; surgeon ergonomic strain. Prevent by careful fascial closure and case selection.

Recent advances

- **da Vinci SP** with same-day discharge pathways; **SP retroperitoneal** renal surgery.
- **Transvesical SP** prostatectomy; hidden incision endoscopic surgery (HIdES) in children.
- New single-port platforms; magnetic anchoring and guidance systems (restore triangulation).

Diagram to draw: umbilical multichannel port with articulating instruments crossing at the fascia (chiral) versus a triangulated multiport diamond.

> **Key points to remember:**
>
> - LESS = one incision; LESSCAR (2008) standardised the term.
> - Core problem: lost triangulation; solved by articulating or curved instruments and robots.
> - Same outcomes as laparoscopy with less pain and better cosmesis; more hernias.
> - da Vinci SP (FDA 2018): one 25–27-mm cannula, wristed instruments.

Q2. Holep

Type: LAQ (25 marks) (also answered in 2019-Summer Paper 1, Q3c and 2020-Summer Paper 4, Q3d)

Read as HoLEP (holmium laser enucleation of the prostate) for BPH (benign prostatic hyperplasia).

Introduction

- **HoLEP** removes the whole transition-zone adenoma endoscopically **along the surgical capsule**. It is the endoscopic equivalent of open simple prostatectomy.
- **Evolution:** HoLAP (holmium laser ablation, 1994, slow) → HoLRP (resection, strips removed with a grasper) → **HoLEP (1998)**, made practical by the **tissue morcellator**.
- "Enucleation is enucleation": any energy giving a true **AEEP (anatomic endoscopic enucleation of the prostate)** gives durable results.

Laser physics

- **LASER:** light amplification by stimulated emission of radiation. Ho:YAG = holmium atoms in a YAG (yttrium–aluminium–garnet) crystal.
- **Wavelength about 2120 nm**; chromophore **water**; **pulsed**; optical penetration **0.4 mm**. It cuts by vaporisation with a thin coagulation zone; heat spreads 2–3 mm and seals small and medium vessels.
- Flexible quartz fibres; **normal saline**, so no TUR (transurethral resection) syndrome.

Laser	Wavelength	Chromophore	Penetration
Holmium:YAG	2120 nm	Water	0.4 mm
Thulium fibre	1940 nm	Water	0.2 mm
KTP (potassium-titanyl-phosphate, green light)	532 nm	Haemoglobin	0.8 mm
Nd:YAG	1064 nm	–	About 1 cm (sloughing)

MOSES (pulse modulation)

- **Moses effect:** the first part of each pulse boils a **vapour channel** in the water. Vapour absorbs the light far less, so the rest of the pulse passes through to the tissue ("parting the water").
- **Moses technology** splits each pulse to exploit this: more energy reaches tissue, and it **may shorten operating time**.

Indications

- **Size-independent** (<30, 30–80, >80 mL), with or without a median lobe (AUA [American Urological Association] Statement 38). Only enucleation spans every size band.
- **High bleeding risk or anticoagulated men** (AUA Statement 43).
- Large glands otherwise needing open prostatectomy; retention; failed medical therapy.

Technique

- **Equipment:** 60–150 W Ho:YAG; **550 µm end-firing fibre**; 24–28 Fr continuous-flow laser resectoscope with 30° lens; saline.
- **Settings:** 2.0 J × 40–50 Hz to enucleate; 1.0–2.0 J × 20 Hz to coagulate.
- **Three-lobe (bottom-up, early apical release):**
  1. Median lobe: **bladder-neck incisions at 5 and 7 o'clock** to the capsule, joined proximal to the verumontanum; median lobe enucleated into the bladder. No median lobe: one 6 o'clock incision.

2. **Perimontanal inverted-U incision**; scope pushes laterally to open the lobe–capsule plane at the apex.
3. Plane carried anteriorly to 11–1 o'clock; **apical mucosal strip incised early** to avoid sphincter traction.
4. Anterior commissure divided; lateral lobes enucleated into the bladder.
5. **Haemostasis** with a defocused beam.

- **Two-lobe:** 6 and 12 o'clock grooves; each half enucleated with its share of median lobe.
- **En bloc:** whole adenoma in one piece after circumferential apical release; usually paired with early apical release.
- **Morcellation:** nephroscope, two open inflow lines, **full bladder** and good haemostasis. Morcellators: Piranha (rotating), Versacut (oscillating). Fragments sent for histology.
- **Catheter:** 22 Fr three-way; same-day or next-day removal.

Outcomes and trials

- **Versus TURP (transurethral resection of the prostate):** RCT (randomised controlled trial) in glands <100 mL: better symptom score and PVR (post-void residual), less haemoglobin drop, shorter catheter time and stay, longer operation.
- **Meta-analyses:** better **IPSS (International Prostate Symptom Score) and Qmax (maximum flow rate)**, durable to **3–8 years**. **HoLEP is the only endoscopic procedure with efficacy superior to TURP.**
- **Versus open prostatectomy** (Naspro 2006; Kuntz 2008, 5 years): equal function, less blood loss, shorter catheter time and stay.
- **Retreatment rare**: whole transition zone removed.
- **Anticoagulated men:** feasible; slightly longer stay; warfarin worse than DOACs (direct oral anticoagulants).
- **Guidelines:** EAU (European Association of Urology) and AUA list HoLEP as a first-line alternative to TURP and to open surgery for large glands.

Complications

- **Intraoperative:** capsular perforation; ureteric orifice injury (glands >80 mL); **morcellator bladder injury** (stop and drain; intraperitoneal → repair); retained adenoma.
- **Early:** transient storage symptoms and dysuria; **transient SUI (stress urinary incontinence)** from adenoma volume removed, falling over 6 months; persistent SUI rare.
- **Late: retrograde ejaculation about 75%**; BNC (bladder neck contracture) **0–3.2%** (small glands <40 mL, consider bladder-neck incision); meatal stricture.
- Erectile function unchanged. Transfusion rare; conversion to TURP almost never.

Learning curve

- The main barrier: from 20 supervised cases; **efficiency plateaus after about 50 cases**. Mentoring and simulators help.

Recent advances

- High-power pulse modulation; **ThuLEP** (thulium laser enucleation) and bipolar enucleation.
- **Ejaculation-preserving and median-lobe-only HoLEP** (antegrade ejaculation about 90%).
- **Same-day discharge** and catheter removal.

Diagram to draw: sagittal prostate showing the enucleation plane along the surgical capsule, with 5 and 7 o'clock incisions.

Key points to remember:

- Ho:YAG 2120 nm, water, 0.4 mm; saline, no TUR syndrome.
- Size-independent; preferred in anticoagulated men.
- Only endoscopic method superior to TURP; retreatment rare.
- Retrograde ejaculation about 75%; learning curve about 50 cases.

Q3a. Neurovascular bundle preservation in Radical prostatectomy: recent techniques

Type: SAQ (10 marks)

Anatomy

- **NVB (neurovascular bundle) of Walsh** (Walsh and Donker, 1982): capsular vessels plus **cavernous nerves** (parasympathetic, from the pelvic plexus). It lies posterolateral, in the lateral pelvic fascia between **prostatic and levator fascia**.
- Nerves are invisible; the **vessels mark them**. Distribution varies (posterolateral bundle, posterior "hammock", anterolateral fibres).
- **Injury mechanisms:** incision, entrapment (clip or suture), **thermal**, **traction**.

Dissection planes

Plane	Where	Use
Intrafascial	Between capsule and prostatic fascia	Maximal nerves; more positive margins
Interfascial	Between prostatic and levator fascia	**Standard nerve-sparing plane**
Extrafascial	Outside levator fascia	Widest margin; erectile dysfunction likely

- Chapter 160 (Walsh) calls intrafascial "never used"; Chapter 161 accepts it in selected men.

Techniques

- **Walsh anatomic (open):** release apex to base from the rectal surface; small clips; **no thermal energy**.
- **High anterior release** (open) and **Veil of Aphrodite** (robotic): lateral prostatic fascia lifted high anteriorly as a "veil". Select by **Tsuzuki criteria**: ≥2 of PSA (prostate-specific antigen) >10 ng/mL, Gleason >6, abnormal DRE (digital rectal examination), perineural invasion, high core involvement → >10% risk, so avoid.
- **Antegrade release (robotic):** incise levator fascia, sweep NVB posterolaterally, clip pedicle at the "groove".
- **Athermal, traction-free:** clips or bulldogs, cold scissors.
- **Graded (risk-stratified) nerve sparing:** full, partial or none per side, from MRI (magnetic resonance imaging) and nomograms.
- **NeuroSAFE:** frozen section of the whole posterolateral margin; if positive, that NVB is resected. More nerve sparing without more positive margins; the NeuroSAFE PROOF RCT (randomised controlled trial) tested its effect on erectile function.
- **Retzius-sparing (Bocciardi 2010)** and **Hood (Tewari 2021):** aimed at continence; potency equal at 12 months.

Adjuncts

- **Penile rehabilitation:** PDE5i (phosphodiesterase type 5 inhibitor), then injections.
- **Also know:** sural nerve grafts and nerve stimulation tests (abandoned); amniotic membrane wraps, fluorescence nerve imaging (experimental).

Diagram to draw: axial prostate showing three fascial planes and NVBs at 5 and 7 o'clock.

Key points to remember:

- Interfascial = standard plane.
- Avoid heat and traction; clips, cold scissors.
- Veil and high release for low-risk sides (Tsuzuki).
- NeuroSAFE frozen section: more sparing, safe margins.

Q3b. Muscle assisted urethroplasty

Type: SAQ (10 marks)

Read as muscle-flap-assisted graft urethroplasty: a vascularised muscle flap used as bed or cover for a graft. "Muscle-sparing" bulbar urethroplasty is noted at the end.

Rationale

- A free graft (e.g. **BMG [buccal mucosal graft]**) has no blood supply. It survives by **imbibition** (about 48 h, nutrients from the bed), then **inosculation** (new microcirculation). So **"a graft lives on its bed"**.
- **Poor beds:** radiation, multiple failed repairs, fistula, pelvic-fracture scar, infection. Here a **muscle flap brings blood supply**, fills dead space and separates suture lines.

Muscle flaps used

Flap	Supply	Use
Gracilis (muscle-only, axial, multiple pedicles)	Medial circumflex femoral branch, proximal thigh	Irradiated posterior urethra, rectourethral fistula, complex perineal repair
Bulbospongiosus	Perineal branches	Covers a ventral graft; support in bulbar repair
Rectus abdominis (VRAM, vertical rectus abdominis myocutaneous)	Inferior epigastric	Large pelvic or perineal defects

Technique (gracilis-assisted BMG)

1. Lithotomy; perineal exposure; excise scar or fistula.
2. Close rectal side; urethra opened.
3. Gracilis harvested through a medial thigh incision; pedicle preserved; tunnelled to perineum.
4. Muscle interposed between rectum and urethra, or used as the graft bed.
5. **BMG quilted onto the muscle** as onlay; urethra closed over a catheter.
6. Catheter drainage; urethrogram before removal.

Results (Chapter 86)

- Gracilis with or without BMG for rectourethral fistula: **non-irradiated, all closed**, no later strictures. **Irradiated:** most closed in one stage, but more strictures and permanent bowel diversion.
- Muscle interposition is integral; BMG is invaluable for urethral closure.

Complications

- Thigh wound problems; graft contraction, restricture, fistula recurrence.

"Muscle-sparing" bulbar urethroplasty (Barbagli)

- The **bulbospongiosus muscle, central perineal tendon and perineal nerves** are left intact; dorsal BMG onlay through a lateral approach.
- Aim: less post-void dribbling and ejaculatory dysfunction.

Diagram to draw: gracilis flap tunnelled to the perineum, lying between rectum and urethra with BMG on it.

Key points to remember:

- Graft survival: imbibition then inosculation.
- Muscle flap rescues a poor (irradiated) bed.
- Gracilis is axial, robust, easy to tunnel.
- Muscle-sparing: preserves bulbospongiosus and nerves.

Q3c. Penile implants.

Type: SAQ (10 marks) (also answered in 2018-Summer Paper 2, Q3d and 2020-Summer Paper 2, Q2)

Indications

- ED (erectile dysfunction) failing, contraindicated for, or declining PDE5Is (phosphodiesterase type 5 inhibitors), intracavernosal injections or vacuum devices.
- Post-prostatectomy, diabetes, pelvic radiotherapy; **Peyronie disease with ED**; corporal fibrosis (**ischaemic priapism >36 h**, implant removal).
- **Contraindications:** active infection (urinary, skin, systemic), uncontrolled diabetes, unrealistic expectations, reversible ED.

Types

Type	Examples	Features
Malleable (semirigid) rods	Tactra (nitinol), Genesis (silver)	Simple, cheap, rarely fail; always firm; erode more in insensate men
2-piece inflatable	Ambicor (discontinued)	No reservoir; poor flaccidity
3-piece inflatable	AMS 700 (CX, LGX), Titan	Cylinders, scrotal pump, reservoir; **best rigidity and flaccidity**

Preparation and consent

- Sterile urine; **blood glucose <200 mg/dL on the day**; chlorhexidine–alcohol skin preparation; vancomycin or cephalosporin plus gentamicin.
- **Consent:** the erect penis may be shorter; sensation and ejaculation may reduce; infection or failure may need reoperation.

Surgical steps (3-piece, penoscrotal)

- Catheter → upper scrotal incision → corporotomy between stay sutures → dilate proximally and distally (instruments dorsolateral) → **goalpost test** for crossover → measure → cylinders equal to or 1 cm shorter.
- Pump in a dependent dartos pouch; reservoir in the **space of Retzius**, or **ectopic submuscular** after pelvic surgery (lockout valve prevents autoinflation).
- Infrapubic approach: reservoir placed under vision; dorsal nerve at risk. Activation at 4–6 weeks.

Complications

- **Infection 1–3%** (biofilm): **antibiotic-coated devices halve it** (InhibiZone rifampin–minocycline; hydrophilic Titan dipped). **No-touch technique** <0.5%. Treat by removing **all** parts; **Mulcahy salvage** succeeds in **84%**.
- **Mechanical failure:** repair about **15% at 5 years, 30% at 10**.
- Crural perforation, crossover, urethral injury; erosion; glans necrosis; reservoir herniation.

Outcomes

- **Satisfaction 80–90%** (patient and partner), highest of any ED treatment.

Diagram to draw: 3-piece inflatable implant: cylinders, scrotal pump, reservoir in the space of Retzius.

Key points to remember:

- 3-piece inflatable: best rigidity and flaccidity.
- Glucose <200 mg/dL; coated device; no-touch.
- Infected implant: remove all; Mulcahy salvage 84%.
- Repair 15% at 5 years; satisfaction 80–90%.

Q3d. Female kidney donors: enumerate special precautions

Type: SAQ (10 marks)

Introduction

- Most living donors, especially in India, are **women** (wives, mothers). Guidance: **KDIGO (Kidney Disease: Improving Global Outcomes) 2017** living donor guideline.

Before donation

1. **Free, uncoerced consent:** interview the donor alone; independent donor advocate; family pressure and gender inequity screened. In India the THOA (Transplantation of Human Organs Act) authorisation committee verifies this.
2. **Pregnancy:** pregnancy test before contrast CT (computed tomography) and surgery; **reliable contraception** during work-up. Pregnant women do not donate.
3. **Obstetric history:** previous **pre-eclampsia, gestational hypertension or gestational diabetes** signals higher future hypertension, diabetes and kidney risk; evaluate and counsel.
4. **Future pregnancy counselling (KDIGO):** donation **increases gestational hypertension and pre-eclampsia** (Garg et al., NEJM 2015: about 11% vs 5% in non-donors). Maternal and fetal outcomes are otherwise similar (Chapter 91).
5. **Oestrogen therapy:** combined OCPs (oral contraceptive pills) and HRT (hormone replacement therapy) raise **VTE (venous thromboembolism)** risk; stop them some weeks before surgery and give barrier or progestogen-only contraception; VTE prophylaxis.
6. **Gynaecological and cancer screening:** cervical cytology, breast examination or mammography (age-appropriate); pelvic ultrasound if indicated.
7. **Urine tests:** repeat urinalysis **outside menstruation** to avoid false haematuria; exclude recurrent UTI (urinary tract infection).
8. **GFR (glomerular filtration rate):** low muscle mass makes creatinine-based estimates unreliable; **measured GFR** preferred. Note nephron mismatch with a large recipient.
9. Correct **iron-deficiency anaemia**.

Operative

- Laparoscopic donor nephrectomy; **Pfannenstiel** extraction for cosmesis; transvaginal extraction possible.

After donation

- **Defer pregnancy** until recovery and stable function; many programmes advise at least 6–12 months.
- In pregnancy: inform obstetrician; **high-risk antenatal care**; monitor BP (blood pressure), proteinuria and creatinine; low-dose aspirin may be considered.
- **Lifelong annual BP, creatinine and proteinuria**; avoid NSAIDs (non-steroidal anti-inflammatory drugs).

Diagram to draw: flowchart: consent → pregnancy test → hormone review → GFR → surgery → pregnancy follow-up.

> **Key points to remember:**
>
> - Ensure free consent (Indian gender gap).
> - Pregnancy test; contraception; stop oestrogens.
> - Donors: gestational hypertension or pre-eclampsia about 11% vs 5%.
> - Defer pregnancy; high-risk antenatal follow-up.

Q3e. Hospital authorization committee for live related kidney donation.

Type: SAQ (10 marks)

Indian law; not covered in the chapter notes.

Legal framework

- **THOA (Transplantation of Human Organs Act) 1994**; amended 2011 as THOTA (Transplantation of Human Organs and Tissues Act); **Rules 2014**.
- Purpose: **prevent organ trade** while allowing genuine donation.

Donor categories

Donor	Approval
Near relative: spouse, son, daughter, father, mother, brother, sister, grandparents, grandchildren	Verified by the hospital's competent authority
Other than near relative (affection or attachment)	**Authorisation Committee approval mandatory**
Swap (paired) donation between near-relative pairs (2011)	Authorisation Committee
Foreign donor or recipient	Authorisation Committee; embassy certifies relationship

Constitution

- **Hospital-based** if a centre does more than **25 transplants a year**; otherwise **district or state committee**.
- **Members:** medical director or superintendent (chair); two senior doctors of the hospital **not in the transplant team**; two persons of high integrity and social standing (preferably one woman), nominated by the State; state committees add health-department nominees.

Functions and procedure

1. Receive the **joint donor–recipient application** with consent forms, fitness certificates and identity and relationship documents.
2. **Personal interview** of donor, recipient and families; proceedings **video-recorded**.
3. Verify **relationship** (documents; **HLA [human leukocyte antigen] or DNA testing** if doubtful).
4. Exclude **commercial dealing**: financial status, reasons, long association, middlemen.
5. Confirm donor understands risks and consents freely, without coercion.
6. Check **domicile** (other-state donors).
7. Decide promptly; decision recorded and displayed; reasons given if refused. **Appeal** to the State Government within **30 days**.

Restrictions

- **Minors and mentally ill people** not accepted as living donors except in exceptional, approved cases.
- **Foreign recipient with unrelated Indian donor** not approved.
- **Penalty** for commercial dealing: imprisonment up to 10 years and fine up to ₹1 crore; doctors' registration removed.

Diagram to draw: flowchart: application → near relative (competent authority) or other (committee) → interview, verify → decision → appeal.

Key points to remember:

- THOA 1994; amended 2011; Rules 2014.
- Hospital-based committee if >25 transplants a year; no transplant team member.
- Non-relatives, swap and foreigners need committee approval.
- Interview video-recorded; appeal within 30 days.

Q3f. Diet in stone disease.

Type: SAQ (10 marks)

Principle

- Diet lowers urinary **supersaturation** by **dilution**, by reducing promoters (calcium, oxalate, uric acid, sodium) and by raising inhibitors (citrate).
- General advice for all; recurrent formers tailored to **24-hour urine**.

General measures

Measure	Target	Mechanism and evidence
Fluid	Urine **≥2.5 L/day** (about 3 L intake), spread through the day	Most important single factor. **Borghi 1996 RCT (randomised controlled trial):** recurrence 12.1% vs 27% at 5 years
Normal calcium	1000–1200 mg/day, with meals	Binds oxalate in the gut. **Restriction increases stones**. **Borghi 2002:** normal-calcium, low-salt, low-protein diet halved stone events vs low-calcium diet
Low sodium	Salt about 4–5 g/day	Sodium increases urinary calcium, lowers citrate
Limit animal protein	0.8–1.0 g/kg/day	Acid and purine load: hypercalciuria, hypocitraturia, hyperuricosuria
Fruit and vegetables (DASH [Dietary Approaches to Stop Hypertension] diet)	Daily	Alkali and citrate; DASH linked to fewer stones
Weight	BMI (body mass index) 18–25	Obesity: low urine pH, more oxalate and uric acid

- **Citrus:** lemonade (most evidence) and orange juice raise citrate; orange juice adds alkali. **Volume matters more than drink type.**
- Avoid sugar-sweetened drinks and **high-dose vitamin C** (converted to oxalate).

Stone-specific diet

Stone or abnormality	Diet
Hyperoxaluria	Avoid large portions of **spinach, beets, chocolate, nuts, tea**; calcium with meals
Enteric hyperoxaluria	Strict low oxalate and **low fat**; **calcium up to 1 g/day with meals**; fluid; magnesium
Uric acid	Less animal protein (purines give about 30% of urinary uric acid); fruit and vegetables; weight loss
Cystine	Fluid **3–5 L/day** including night; sodium **≤2300 mg/day**; less animal protein (not in children)
Struvite	Infection control; diet has little role

Adherence

- Poor: only half reach 2.5 L at 6 months; apps and smart bottles help.

Diagram to draw: "stone-prevention plate": water, fruit and vegetables, normal dairy, little meat and salt.

Key points to remember:

- Urine ≥2.5 L/day (Borghi 1996).
- Never restrict calcium (Borghi 2002).
- Low salt, less animal protein, more fruit and vegetables.
- Cystine: ≥3 L urine; enteric: low fat, calcium with meals.

Rapid Revision Sheet – 2023-Winter Paper 4

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Laproendoscopic single site surgery: LESS (LAQ)

- LESS = one incision; LESSCAR (2008) standardised the term.
- Core problem: lost triangulation; solved by articulating or curved instruments and robots.
- Same outcomes as laparoscopy with less pain and better cosmesis; more hernias.
- da Vinci SP (FDA 2018): one 25–27-mm cannula, wristed instruments.

Q2. Holep (LAQ)

- Ho:YAG 2120 nm, water, 0.4 mm; saline, no TUR syndrome.
- Size-independent; preferred in anticoagulated men.
- Only endoscopic method superior to TURP; retreatment rare.
- Retrograde ejaculation about 75%; learning curve about 50 cases.

Q3a. Neurovascular bundle preservation in Radical prostatectomy: recent techniques (SAQ)

- Interfascial = standard plane.
- Avoid heat and traction; clips, cold scissors.
- Veil and high release for low-risk sides (Tsuzuki).
- NeuroSAFE frozen section: more sparing, safe margins.

Q3b. Muscle assisted urethroplasty (SAQ)

- Graft survival: imbibition then inosculation.
- Muscle flap rescues a poor (irradiated) bed.
- Gracilis is axial, robust, easy to tunnel.
- Muscle-sparing: preserves bulbospongiosus and nerves.

Q3c. Penile implants. (SAQ)

- 3-piece inflatable: best rigidity and flaccidity.
- Glucose <200 mg/dL; coated device; no-touch.
- Infected implant: remove all; Mulcahy salvage 84%.
- Repair 15% at 5 years; satisfaction 80–90%.

Q3d. Female kidney donors: enumerate special precautions (SAQ)

- Ensure free consent (Indian gender gap).
- Pregnancy test; contraception; stop oestrogens.
- Donors: gestational hypertension or pre-eclampsia about 11% vs 5%.
- Defer pregnancy; high-risk antenatal follow-up.

Q3e. Hospital authorization committee for live related kidney donation. (SAQ)

- THOA 1994; amended 2011; Rules 2014.
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Q3f. Diet in stone disease. (SAQ)

- Urine ≥2.5 L/day (Borghi 1996).
- Never restrict calcium (Borghi 2002).
- Low salt, less animal protein, more fruit and vegetables.
- Cystine: ≥3 L urine; enteric: low fat, calcium with meals.